



Vegakits plastic body robot

4. Professional colleges
5. Entertainment industry

Building blocks

Robotic kits are great integrators as they give insights into the fields of mechanics, electronics as well as programming. Most kits are intended for easy assembly. Some kits come with their own software, while some use commonly available software.

Robotic hardware kits can be mechanically configured to enable different types of manipulation and locomotion. Electronic modules can be used to control the mechanics through actuators as well as to gather information from the hardware and the environment with the help of sensors. Software can be used to add behaviour and intelligence.

The kit may include the brain of the robot, motors, different types of

sensors (light, sound, bump, distance, etc) and mechanical parts like links, wheels and gears. It can be used to create different intelligent machines, be it an obstacle avoider, automatic crane or garbage collector.

The mechanical structure of the kit acts as a chassis or body of the robot. The electronic controller board controls all the electromechanical parts and inherits the logic part of the robot. Sensors/actuators and other interface devices help the robot to connect to the world, by sensing it and then taking action as a response to the stimulus. Sensors use transducers to change the input signal (sound, light, pressure, temperature, etc) into an analogue or digital form capable of being used by the robot.

A variety of electric motors provide power to robots, allowing them to move material, parts, tools or special-

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ised devices with various programmed motions.

Generally, two types of batteries are used to provide power supply: Primary batteries are used once and then discarded. Secondary batteries operate from a (mostly) reversible chemical reaction and can be recharged several times.

Software tools come with application software to configure, sequence and also develop and download algorithms into the robot. High-end software tools do simulation and 3D visualisation to assist the entire robot development life-cycle.

Microcontrollers (MCUs) are intelligent electronic devices used inside robots. A microcontroller takes input from the device and controls it by sending signals to different components

“Many vendors are selling cheap, unbranded robotic items.

Many Chinese items have also come into the market. These vendors have no knowledge of robotics.”—Ajit Vira, general manager, MacNet Technology



in the device. Robotic kits use various platforms such as 8051, AVR and even 32-bit ARM processor. These perform various functions ranging from path planning, vision-based processing to Swarm behavioural functions. For programming, generally, any ‘C’ compiler

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is preferred.

The LEGO Mindstorms platform comes with the LEGO plug-and-play guarantee. LEGO parts are used for the assembly and creation of robots. The LEGO controller is an ARM-based controller that can be programmed using a USB or a Bluetooth connection. Many different kinds of environments are available for programming, including LabView, C, C++ and Java. Support

Components of a standard kit

1. Programmable platform (the logic processor of the bot)
2. Motors (actuators to move the bot)
3. Sensors (to gather information about surroundings)
4. Power supply
5. PC interface kit (to enable programming of the bot via PC)



Boe-Bot robot kit from rhydoLABZ